

A Survey on Deep Learning Techniques for Joint Named Entities and Relation Extraction

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Abstract—Named Entity Recognition (NER) and Relation Extraction (RE) are two principal subtasks of knowledge-based systems that extract meaningful information from unstructured text. With Recent advances in Deep Learning techniques, new models use Joint Named Entities and Relation Extraction (JNERE) techniques that simultaneously accomplish NER and RE subtasks. These models avoid the drawbacks of using the traditional pipeline method. As contributions of our study to the other related works, we specifically survey JNERE techniques. The reason for not focusing on pipeline methods or other older techniques is the recent advances of JNERE methods in achieving the state-of-art results for most databases. Additionally, we provide a comprehensive report on the embedding techniques and datasets available for this task. Finally, we discuss the approaches and how they improved the results.

Index Terms—Name Entity Recognition, Joint Relation Extraction, Knowledge-based Systems, Deep Learning

I. INTRODUCTION

Information Extraction (IE) is the core of knowledge-based systems that utilizes Natural Language Processing (NLP) techniques to discover hidden information of unstructured texts [1] [2]. Recently there are many knowledge-based systems available to assist experts in doing their jobs like [3] for PC troubleshooting and [4] that has medical application to Diagnose Diabetes. IE's two significant subtasks are Named Entity Recognition (NER) and Relation Extraction (RE). Firstly, NER is the task of detecting all regularly named entities such as the name of locations, organizations, persons, or unique entities like the name of drugs, chemical components, biological proteins, etc. NER is the primary step of shaping a knowledge-based system [5]. Secondly, the RE task is to identify relations between detected named entities in the text. The RE system should find all possible triples in the form of (Entity 1, Relation type, Entity 2). Most papers researching RE call entity 1 "subject" and entity 2 "object." Therefore, in this paper, we show the output of the RE system all triples of the text like (subject, relation, object).

According to the relation types, studies in RE are classified into two main categories: Open IE [6] and a predefined set of relations [1]. Open IE methods are not restricted to finding a small group of target relations that are already defined. On the contrary, it can improve the system to see beyond the

specified classes of relation. However, this paper focuses on studies with a predefined set of relations.

There are two types of RE models with a predefined set of relation classes, Joint Named Entity Relation Extraction (JNERE) and pipeline methods. The pipeline techniques such as [7], trying to find all possible entity pairs available in the text in the first phase and then attempt to classify the relations of the pairs in the second phase. However, there are a lot of drawbacks accompanying this pipeline method that we describe in the following points:

- Error in the NER subtask can propagate to the RE subtask. For example, if the system misidentifies a named entity, the approach misassigns it to a class of relation.
- There will be a highly imbalanced classes of relations because there are many pairs available in texts that do not belong to any relation class. Therefore, the false class, which is the class of not having a relation, will get the most members.
- Uncovering a massive number of entity pairs increases the complexity of the problem.
- The classifier will be confused when the same subject and object belong to more than one relation classes.

Basically, RE sequences are divided into three categories: Normal, Single Entity Overlapped (SEO), and Entity pair Overlapped (EPO). SEO sequences have at least two triples with subject or object in common. EPO sequences have at least two triples with the same subject and objects but different relation class. Normal sequences do not have any triples with subject and object in common [8]. Another concern of the pipeline method in RE is SEO and EPO sequences.

To alleviate the mentioned challenges of the pipeline method, JNERE was proposed. Most of the papers in this field named joint named entities relation extraction, triple extraction, and end-to-end relation extraction. All refer to approaches that simultaneously extract entities and relations available in the texts.

II. RELATED WORK

The survey paper [9] categorized the methods for RE task using Deep Neural Networks into supervised, semi-

supervised, and unsupervised methods. This study explicitly reviews pipeline techniques and skips JNERE methods.

The study [10] [11] concentrates on techniques using Distant supervision and organize them based on their approaches to noise reduction, embedding, and leveraging auxiliary information techniques. The paper reviews methods that mitigate the effect of noisy instances in data.

E.Shahab et al. [12] and [13] focus on RE in biomedical application [14] and methods that classify biomedical named entities to Gene-Disease, Protein-Protein, Gene-Protein, etc. relations.

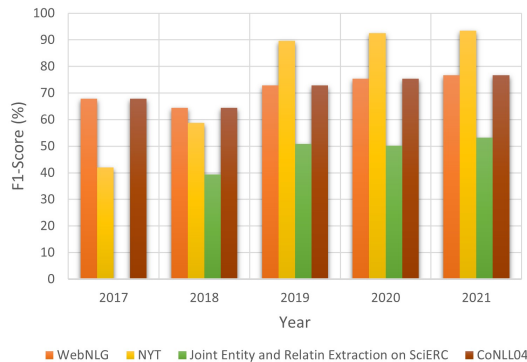


Fig. 1. Comparison of reported F1-score of different methods using JNERE methods in recent five years on four famous NER datasets.

One of the papers that reviews a few joint representation learning models as a part of the study is [15] which was done in 2017, and since then, many models have been proposed that achieved considerable success in JNERE.

The most recent paper is [10] that firstly surveys many NER techniques for English, multiple languages in domain-specific databases. Secondly, the article explains the RE techniques and categorizes them into neural network-based and non-neural methods. The paper mainly concentrates on pipeline techniques. The following points are the main contributions of our paper.

Figure 1 illustrates how JNERE methods progressed on the four primary databases during the last five years. This figure indicates the importance of studying JNERE models. Therefore, as the contributions of our paper to the related works, we specifically classify advanced methods for the JNERE task. We categorize the JNERE model's word embedding techniques into feature-based models and Neural Network (NN) based models and explain the most recent proposed JNERE methods based on their deep learning (DL) techniques. Additionally, we provide statistical information and charts about NER databases and discuss methods that achieve state-of-art results.

III. DATASETS FOR NAMED ENTITY RELATION EXTRACTION

This section provides information about supervised datasets for the NER task. A NER corpus contains many

texts whose named entities are tagged, and a set of pre-defined relations are defined and allocated to the pairs of entities available in the documents. Table 1 summarizes some widely used datasets in the proposed methods. It provides information about, the datasets' name, description, number of relations classes, number of documents, number of sentences, being sentence or document level, and text type.

The first section of the table is for biomedical datasets (GENIA [16], BioInfer [17], ADE [18], DDI [19], CHEMPROT [20], DrugProt [21], WLPC [22]) to extract biomedical named entities such as drugs, protein, chemical products, etc., from texts. The relation extraction assignment is essential in bioinformatic since many challenges have tracks for this task. For example, the DDI (Drug-Drug Interaction) dataset is from the 2011 and 2013 DDIExtraction challenges, and the CHEMPROT and DrugProt datasets are from BioCreative challenges in 2017 and 2021, respectively. The middle of the table is for news-based datasets (CoNLL04 [23], ACE04 [24], NYT10, NYT11, NYT24, NYT29 [25]). A few of these datasets are multilingual, such as ACE04 and ACE05, in addition to English, including Chinese and Arabic data.

The last section of the table consists of datasets from Wiki data such as WikiReading [26], Wiki-KBP [27], and DocRED [28]. The abstract of the scientific papers is a good resource for datasets. Therefore, there are datasets from scientific resources such as SemEval 2010 [29].

A. Document-level and Sentence-level Datasets

Based on the position of the relations and pairs of entities in the text, datasets are divided into sentence-level and document-level. At the sentence-level datasets, both entities and their relationship belong to one sentence. However, in document-level datasets, some entities with relationships might be cross sentences, with a distance of more than one sentence's length. Therefore, in a document-level dataset, if the method separates the text based on sentences, some relations will be dropped. Most of the datasets in table 1 are sentence-level. However, a few datasets, such as GENIA, DocRED, WikiReading, and SciERC [30], are document-level, and CHEMPROT and DrugProt have a few relationships across the documents.

IV. WORD EMBEDDING TECHNIQUES

Word embedding is the JNERE systems' first step that converts the input text into model understanding format. It grabs the syntactic and semantic information of the vocabularies. The JNERE methods based on their primary word embedding techniques are divided into feature-based models and Neural network-based methods. In the following section, we describe models in these two groups.

A. Feature-based Methods

The idea of feature-based models is like word position techniques which is the words closer to the target entities include more helpful information for the relation classification task between entities. For example, research [34] [35]

TABLE I
LIST OF THE DATASETS FOR JOINT ENTITIES RELATION EXTRACTION TASK

Corpus	Description	# Relation Classes	# of Documents	# of Sentences	Sentence Level	Document level	Text Typ			
							Biological	News	Paper abstract	Wiki
GENIA [16]	Medline abstracts	47	2000	18,545	•	•	•	-	•	-
BioInfer [17]	Abstract of Biomedical Research	5	-	1,100	•	-	•	-	•	-
ADE [18]	Biomedical Data (Adverse Drug Event)	2	3000	20,967	•	-	•	-	-	-
DDI [19]	drug- drug interactions from biomedical texts	2	-	33,553	•	-	•	-	-	-
CHEMPROT [20]	BioCreative VI chemical- protein interaction Track 5	6	2420	-	•	•	•	-	•	-
DrugProt [21]	BioCreative VII chemical- protein interaction Track 1	13	5,000	-	•	•	•	-	•	-
WLPC [22]	Web lab protocols	12	-	13,679	•	-	•	-	-	-
CoNLL04 [23]	Newspaper	5	-	1,441	•	-	-	•	-	-
ACE04 [24]	English, Arabic, and Chinese News-wire and online forums	6	-	8.7k	•	-	-	-	-	-
ACE05 [31]	English, Arabic, and Chinese News-wire and online forums	6	-	14.5k	•	-	-	•	-	-
NYT10 [25]	New York Times newspaper	52	-	627,827	•	-	-	•	-	-
NYT11 [25]	New York Times newspaper	24	-	337,295	•	-	-	•	-	-
NYT24 [25]	New York Times newspaper	24	-	61,196	•	-	-	•	-	-
NYT29 [25]	New York Times newspaper	29	-	67,312	•	-	-	•	-	-
SemEval 2010 [29]	Sciences and, Physics publications	18	-	10,717	•	-	-	-	•	-
SciERC [30]	Abstract of computer science research papers	7	500	2.7k	-	•	-	-	•	-
WikiReading [26]	Wikipedia	884	18.58M	-	-	•	-	-	-	•
Wiki-KBP [27]	Wikipedia articles	19	780,549	1.51M	•	-	-	-	-	•
DocRED [28]	Wikipedia and Wikidata	96	5,053	-	-	•	-	-	-	•
WebNLG [32]	DBpedia [33] categories	216	6,222	-	•	-	-	-	-	•

handle the word representation in a vector using part-of-speech (POS) tags, Dependency Tree (DT), an entity labels. Factor-based Compositional Embedding Model (FCM) [36] applies dependency trees and named entities from sentence-level to shape word embedding vector. P. Gupta et al. [37] concatenate 50-dimensional word embedding with POS, capital features, and 7-gram [38] with piggybacked entity vectors to shape the final word embedding vector, which will be shared and updated during the training of the RE task. The accuracy of the models using Feature-based methods depends on feature engineering techniques and the NLP tools that extract the features. Therefore, recent studies use Neural Network (NN) techniques to extract the features [39].

B. Neural network-based Methods

Many research using pre-trained models or LSTM, CNN models to capture the features of the text at the character or word level to generate the embedding vector. Papers [40] [41] [42] [43] are NN-based models using Bidirectional Encoder Representation from Transformers (BERT) [44] [45] [46] [47]–[52] pre-trained model to represent words. The encoder attempts to represent the vector of every single word in the

sequence contextually. BERT is a stack of transformers with the following formula:

$$I_{word} = E_{token} + E_{position} \quad (1)$$

I_{word} is the word input of the encoder. E_{token} is the token created by Word Piece embedding dictionary of size 30,000 words, and $E_{position}$ presents the position index of the word in the sequence. Therefore, the embedding vector of the BERT has prior knowledge of the word and its position in the sentence [53].

Elmo is another popular pre-trained model used in NERE systems that employ the bidirectional language model (biLM). It utilizes all layers of the deep pre-trained neural network to represent the word vector in the context of the entire text. Moreover, it is character-based, which uses morphological hints in the word representation that works beyond the vocabulary token unseen in the training data. Research [54] provided different models of Elmo trained on various datasets. Studies conducted by [55] trained this model on clinical domains such as Wikipedia pages for clinical ontology called SNOMED CT [56] and discharge abstract of

radiology writings on MIMIC-III dataset [57]. Some studies run CNN on the character of the input sequence to create the pre-trained embedding representation such as [58].

V. ENTITIES AND RELATIONS LEARNING METHODS

Models mainly utilize different deep learning techniques to do the named entity recognition and relation extraction tasks. For example, available studies employ LSTM, Bi-LSTM (Bidirectional LSTM), CNN, Graph Neural Networks (GNN), or hybrid models in the encoder or decoder sections of figure 2. In the following sections, we classify models based on their learning class. Figure 2 provides an overview of different section of the following methods. Table 2 compares the methods on their experimental dataset and the F1 score metric on relation extraction task.

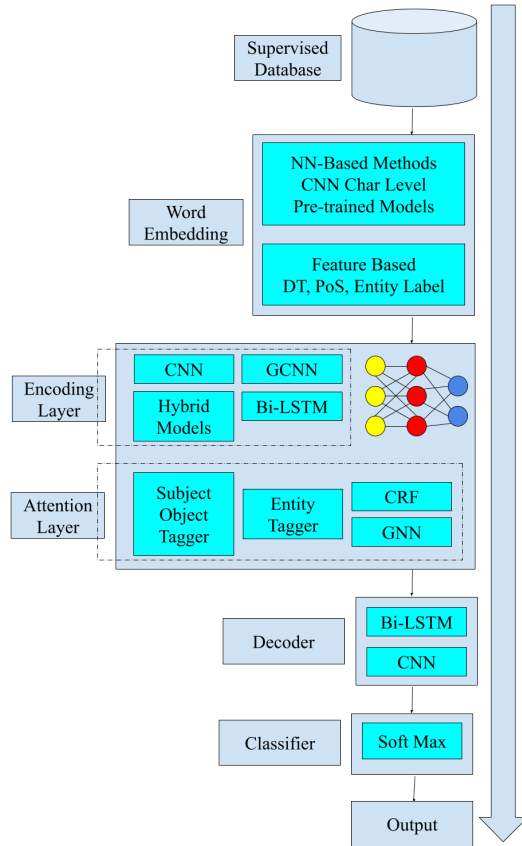


Fig. 2. Overview of Deep Neural Networks-Based Methods

A. LSTM-based methods

Bi-LSTM is the most popular DL method being used in papers [59] [60] [69] [70] [71] to learn the entities and relations of the texts. LSTM is a Recurrent Neural Network with outstanding performance on tasks that require remembering long-term dependencies. There are a lot of studies using LSTM in the JNERE task because, in this job, it is essential to select a model that reflects the structural and grammatical features of the previous words to detect relations through

TABLE II
COMPARING METHODS PERFORMANCE USING DEEP LEARNING TECHNIQUES

Author	Model	Data Set	F1 Score
[59]	Bi-LSTM Relation-Based Attention	NYT	0.846
		WebNLG	0.821
[60]	Bi-LSTM Attention-layer	NYT24	0.844
		NYT29	0.716
[61]	Bi-LSTM Attention-layer CRF layer	DDI	0.702
		CPI	0.49
[62]	Bi-LSTM CRF Multiple tail entities	CoNLL04	0.775
		ADE	0.962
[42]	Bi-LSTM Bidirectional Tree Tagging (BiTT)	NYT	0.711
[63]	Bi-LSTM CRF Attention-layer	ACE04	0.474
		CoNLL04	0.679
		DREC	0.53
		ADE	0.755
[64]	Self-Attention Bi-LSTM	SemEval2010	0.853
[65]	Bi-LSTM Graph Convolutional Networks (GCNs)	WebNLG	0.429
		NYT	0.619
[66]	Heterogeneous GNN Subject and Object Tagger	WebNLG	0.926
		NYT	0.92
[67]	Hybrid Method Bi-LSTM CNN	ACE05	0.842
[68]	Multi-head Self-attention Bi-LSTM CGCNN	SemEval2010	0.851
		CoNLL04	0.879
[13]	GNN and BioBERT	CHEMPROT	0.69
		DrugProt	0.718

long sentences. For example, X. Luo et al. [42] designed a Bidirectional Tree Tagging (BiTT) scheme. BiTT scheme has two phases: relation to Tree and Tree to tags. The first phase converts the relations into trees. It gets the first entity of a sentence as the root of the first Tree called E1 and then moves through the sentence from left to right and adds the entities related to E1. When All entities led by E1 are done, the algorithm selects the first of them as the next root. Finally, forest F is created, and it will be converted into a binary tree. The second phase sets a tag for every word by parsing the binary Tree. The paper is using Bi-LSTM to implement this forward and backward iteration through sentences in Tree format. BiTT handles the challenge of extraction relations from EPO sequences because it is creating different trees for triples with subject or objects in common.

B. Yu. et al. [58] decompose the problem into multi-labeling problems. First, tagging all Head-Entities (HE) and second, Tail Entity and Relation (TER) extraction subtasks. All entities are span based, meaning the entity's start and the end are tagged. The paper utilizes Bi-LSTM to implement the span boundaries predictor.

Some studies use an attention layer or a CRF layer on top of the Bi-LSTM layer to obtain the importance of each word in the sentence. Papers [61] [72] [62] integrate the CRF layer. G. Bekoulis et al. [63] add a CRF vector with

named entities tags in its position, and tag "O" shows words that are not named entities. For example, the CRF layer for the sentence "Smith lives in California." will be (B-PER, O, O, B-Loc, O). B-person and B-Loc respectively illustrate person and location named entity classes. In some reports, attention appears in a binary vector format with a value 1 in the named entities' positions in the sentence [40]. J. Sun et al. [64] utilize multi-head attention, which is a variant of dot-product attention. It provides a query matrix, key matrix, and key-value to the attention layer to compute the dot-product attention.

B. Graph Neural Network-based methods

Recently Graph Neural Network provided a significant application in JNERE task. For instance, research conducted by [65] [73] [43] [13] using this technique. Most of the Graph Convolutional Network (GCN) methods consider each tokenized word of a sentence as a node of a graph and collect contextual information of neighboring nodes. Each node can propagate the information to its adjacent nodes. Paper [74] provides details on how GCN works. In the NERE applications, paper [65] applied a two-phase method. In the first phase, a dependency tree of the sentences is created to shape the adjacency matrix then GCN is applied to extract dependency features of the sentences. Also, in this phase, the model tries to predict the named entities and detect the relations of the entity pairs, which is not the final prediction, and it is considered as a starting point. In the second phase, a relation-weighted bi-GCN is generated, named entities as nodes and relation weight as edges. During the model's training, this phase uses message passing to update the edges and nodes' presentation to improve the predictions of triples.

As another example, K. Zhao et al. [66] use heterogeneous GNN to represent the tokenized word of the sentences. They consider two types of nodes, relations, and words of the sentences; edges only connect relation nodes and word nodes. Then during the training phase, nodes embedding vectors are being updated using iterative fusion message passing techniques. Therefore, words representation already knows their potential relation classes. Finally, the output of the GNN is fed into the tagging method proposed by [40] to detect the named entities boundary and their class of relations.

C. Hybrid Neural Network Methods

Many Methods employ different DL techniques to do the joint NERE task [75]. These techniques are called hybrid methods. For instance, study [67] uses Bi-LSTM cells in the encoder and decoder part of the method to pull the entities, and then there is a CNN layer on top to extract the relations between entity pairs. The contextual information collected from the Bi-LSTM layer is passed to the CNN layer to improve the RE subtask. Z. Meng et al. [68] applied a Multi-head Self-attention (MS) layer in the encoder using Bi-LSTM cells. After text is encoded through this layer, the character level features are extracted using Character Graph Convolutional Network (CGCN). The network consists of

two types of nodes, the Entity-Relationship nodes and the Other nodes. Other nodes indicate the words that are not named entities and tagged as "O" in the dataset. The method starts traversing "O" nodes from left and right. Each word is mapped into a low-dimensional representation space, and it is fed into the CGCN to extract the character features. The method has an outstanding performance in capturing contextual information because it is being done in two layers. M. Miwa et al. [34] designed an end-to-end method in four layers using feature-based embedding. In the first layer, words of the input text are embedded using part of speech tags and dependency type. In the second layer sequence feature of the text is captured using a bidirectional LSTM-RNN unit. The third layer is a single perceptron NN model that gets entity tag, type, and position representation. Finally, the last layer utilizes a bidirectional tree-structured LSTM-RNN unit to assign relations based on the dependency around the target entity pairs.

VI. CONCLUSION

This paper describes the challenges of using traditional pipeline methods and explains a substitution concept called joint named entities relation extraction. We categorized the recent proposed supervised methods in JNERE that are using Deep Neural Networks into three main classes. First, LSTM-based methods, which are mainly an encoder and decoder, sequentially use Bi-LSTM cells to extract the text, entities, and relations contextually. Second, Graph Neural Network-based methods that have various architectures designed to extract the features, entities, and relations included in the text. In this class, some methods consider words of the sequence as the nodes of the graph and mainly use the attention network concept to indicate the importance of words in the sentence. Some methods assume relations as separate node types or assign relations as a weight to the edge of the GNN. Finally, the last class is hybrid methods that employ a combination of DL techniques. Mainly, proposed methods in this class are a mixture of Bi-LSTM with CNN, RNN or GNN. Although JNERE methods make a significant improvement compared to pipeline methods, there are still challenges with Document-level relation extraction which goes beyond one sequence or sentence and endeavors to find relations between long-distance entities.

REFERENCES

- [1] S. Kumar, "A survey of deep learning methods for relation extraction," *arXiv preprint arXiv:1705.03645*, 2017.
- [2] M. E. Z. N. Kambar, P. Nahed, J. R. F. Cacho, G. Lee, J. Cummings, and K. Taghva, "Clinical text classification of alzheimer's drugs' mechanism of action," in *Proceedings of Sixth International Congress on Information and Communication Technology*. Springer, 2022, pp. 513–521.
- [3] A. W. Dahouk and S. S. Abu-Naser, "A Proposed Knowledge Based System for Desktop PC Troubleshooting," 2018.
- [4] I. M. Dheir, A. S. Abu Mettleq, A. A. Elsharif, M. N. Abu Alqumboz, and S. S. Abu-Naser, "Knowledge Based System for Diabetes Diagnosis Using SL5 Object," 2019.
- [5] V. Yadav and S. Bethard, "A survey on recent advances in named entity recognition from deep learning models," *arXiv preprint arXiv:1910.11470*, 2019.

- [6] C. Niklaus, M. Cetto, A. Freitas, and S. Handschuh, "A survey on open information extraction," *arXiv preprint arXiv:1806.05599*, 2018.
- [7] Y. S. Chan and D. Roth, "Exploiting syntactico-semantic structures for relation extraction," in *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, 2011, pp. 551–560.
- [8] Z. Wei, J. Su, Y. Wang, Y. Tian, and Y. Chang, "A Novel Cascade Binary Tagging Framework for Relational Triple Extraction," pp. 1476–1488, 2020.
- [9] H. Wang, G. Lu, J. Yin, and K. Qin, "Relation extraction: A brief survey on deep neural network based methods," in *2021 The 4th International Conference on Software Engineering and Information Management*, 2021, pp. 220–228.
- [10] Z. Nasar, S. W. Jaffry, and M. K. Malik, "Named entity recognition and relation extraction: State-of-the-art," *ACM Computing Surveys (CSUR)*, vol. 54, no. 1, pp. 1–39, 2021.
- [11] A. Smirnova and P. Cudré-Mauroux, "Relation extraction using distant supervision: A survey," *ACM Computing Surveys (CSUR)*, vol. 51, no. 5, pp. 1–35, 2018.
- [12] E. Shahab, "A short survey of biomedical relation extraction techniques," *arXiv preprint arXiv:1707.05850*, 2017.
- [13] M. Esmail Zadeh Nojoo Kambiar, A. Esmailzadeh, and K. Taghva, "Chemical-gene relation extraction with graph neural networks and bert encoder," in *2022 International Conference on Innovations in Computing Research (ICR'22)*, 2022.
- [14] R. Iranzad, X. Liu, W. A. Chaovalitwongse, D. Hippe, S. Wang, J. Han, P. Thammasorn, C. Duan, J. Zeng, and S. Bowen, "Gradient boosted trees for spatial data and its application to medical imaging data," *IJSE Transactions on Healthcare Systems Engineering*, pp. 1–15, 2021.
- [15] S. Pawar, G. K. Palshikar, and P. Bhattacharyya, "Relation extraction: A survey," *arXiv preprint arXiv:1712.05191*, 2017.
- [16] J.-D. Kim, T. Ohta, Y. Tateisi, and J. Tsujii, "GENIA corpus—a semantically annotated corpus for bio-textmining," *Bioinformatics*, vol. 19, no. suppl_1, pp. i180—i182, 2003.
- [17] S. Pyysalo, F. Ginter, J. Heimonen, J. Björne, J. Boberg, J. Järvinen, and T. Salakoski, "BioInfer: a corpus for information extraction in the biomedical domain," *BMC bioinformatics*, vol. 8, no. 1, pp. 1–24, 2007.
- [18] H. Gurulingappa, A. M. Rajput, A. Roberts, J. Fluck, M. Hofmann-Apitius, and L. Toldo, "Development of a benchmark corpus to support the automatic extraction of drug-related adverse effects from medical case reports," *Journal of biomedical informatics*, vol. 45, no. 5, pp. 885–892, 2012.
- [19] Z. Zhao, Z. Yang, L. Luo, H. Lin, and J. Wang, "Drug drug interaction extraction from biomedical literature using syntax convolutional neural network," *Bioinformatics*, vol. 32, no. 22, pp. 3444–3453, 07 2016. [Online]. Available: <https://doi.org/10.1093/bioinformatics/btw486>
- [20] N. Hailu, L. Hunter, and K. B. Cohen, "UColorado_SOM: extraction of drug-drug interactions from biomedical text using knowledge-rich and knowledge-poor features," in *Second Joint Conference on Lexical and Computational Semantics (* SEM)*, Volume 2: *Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013)*, 2013, pp. 684–688.
- [21] A. Miranda, F. Mehryary, J. Luoma, S. Pyysalo, A. Valencia, and M. Krallinger, "Overview of drugprot biocreative vii track: quality evaluation and large scale text mining of drug-gene/protein relations," in *Proceedings of the seventh BioCreative challenge evaluation workshop*, 2021.
- [22] C. Kulkarni, W. Xu, A. Ritter, and R. Machiraju, "An annotated corpus for machine reading of instructions in wet lab protocols," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*. New Orleans, Louisiana: Association for Computational Linguistics, Jun. 2018, pp. 97–106. [Online]. Available: <https://aclanthology.org/N18-2016>
- [23] D. Roth and W.-t. Yih, "A linear programming formulation for global inference in natural language tasks," ILLINOIS UNIV AT URBANA-CHAMPAIGN DEPT OF COMPUTER SCIENCE, Tech. Rep., 2004.
- [24] A. Mitchell, S. Strassel, S. Huang, and R. Zakhary, "Ace 2004 multilingual training corpus," *Linguistic Data Consortium, Philadelphia*, vol. 1, p. 1, 2005.
- [25] S. Riedel, L. Yao, and A. McCallum, "Modeling relations and their mentions without labeled text," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 2010, pp. 148–163.
- [26] D. Hewlett, A. Lacoste, L. Jones, I. Polosukhin, A. Fandrianto, J. Han, M. Kelcey, and D. Berthelot, "Wikireading: A novel large-scale language understanding task over wikipedia," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Berlin, Germany: Association for Computational Linguistics, Aug. 2016, pp. 1535–1545. [Online]. Available: <https://aclanthology.org/P16-1145>
- [27] J. Ellis, X. Li, K. Griffitt, S. M. Strassel, and J. Wright, "Linguistic Resources for 2013 Knowledge Base Population Evaluations," in *TAC*, 2012.
- [28] Y. Yao, D. Ye, P. Li, X. Han, Y. Lin, Z. Liu, Z. Liu, L. Huang, J. Zhou, and M. Sun, "Docred: A large-scale document-level relation extraction dataset," *arXiv e-prints*, pp. arXiv–1906, 2019.
- [29] K. Gábor, D. Buscaldi, A.-K. Schumann, B. QasemiZadeh, H. Zargayouna, and T. Charnois, "Semeval-2018 task 7: Semantic relation extraction and classification in scientific papers," in *Proceedings of The 12th International Workshop on Semantic Evaluation*, 2018, pp. 679–688.
- [30] Y. Luan, L. He, M. Ostendorf, and H. Hajishirzi, "Multi-task identification of entities, relations, and coreference for scientific knowledge graph construction," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. Brussels, Belgium: Association for Computational Linguistics, Oct.-Nov. 2018, pp. 3219–3232. [Online]. Available: <https://aclanthology.org/D18-1360>
- [31] I. I. Ntroduction, "The ACE 2005 (ACE 05) Evaluation Plan Evaluation of the Detection and Recognition of ACE Entities , Values , Temporal Expressions , Relations , and Events 1."
- [32] C. Gardent, A. Shimorina, S. Narayan, and L. Perez-Beltrachini, "Creating training corpora for nlg micro-planning," in *55th annual meeting of the Association for Computational Linguistics (ACL)*, 2017.
- [33] S. Auer, C. Bizer, G. Kobilarov, J. Lehmann, R. Cyganiak, and Z. Ives, "Dbpedia: A nucleus for a web of open data," in *The semantic web*. Springer, 2007, pp. 722–735.
- [34] M. Miwa and M. Bansal, "End-to-end relation extraction using lstrms on sequences and tree structures," *arXiv preprint arXiv:1601.00770*, 2016.
- [35] X. Ren, Z. Wu, W. He, M. Qu, C. R. Voss, H. Ji, T. F. Abdelzaher, and J. Han, "Cotype: Joint extraction of typed entities and relations with knowledge bases," in *Proceedings of the 26th International Conference on World Wide Web*, 2017, pp. 1015–1024.
- [36] M. Yu, M. Gormley, and M. Dredze, "Factor-based compositional embedding models," in *NIPS Workshop on Learning Semantics*, 2014, pp. 95–101.
- [37] P. Gupta, H. Schütze, and B. Andrassy, "Table filling multi-task recurrent neural network for joint entity and relation extraction," in *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, 2016, pp. 2537–2547.
- [38] A. Esmailzadeh, J. R. Fonseca Cacho, K. Taghva, M. E. Z. N. Kambiar, and M. Hajiali, "Building wikipedia n-grams with apache spark," in *Intelligent Computing*. Springer, 2022.
- [39] S. Khorshidi, G. Mohler, and J. G. Carter, "Assessing gan-based approaches for generative modeling of crime text reports," in *2020 IEEE International Conference on Intelligence and Security Informatics (ISI)*. IEEE, 2020, pp. 1–6.
- [40] Z. Wei, J. Su, Y. Wang, Y. Tian, and Y. Chang, "A novel cascade binary tagging framework for relational triple extraction," *arXiv preprint arXiv:1909.03227*, 2019.
- [41] Y. Shen, X. Ma, Y. Tang, and W. Lu, "A Trigger-Sense Memory Flow Framework for Joint Entity and Relation Extraction," in *Proceedings of the Web Conference 2021*, 2021, pp. 1704–1715.
- [42] X. Luo, W. Liu, M. Ma, and P. Wang, "BiTT: Bidirectional Tree Tagging for Joint Extraction of Overlapping Entities and Relations," *arXiv preprint arXiv:2008.13339*, 2020.
- [43] D. Wadden, U. Wennberg, Y. Luan, and H. Hajishirzi, "Entity, relation, and event extraction with contextualized span representations," *arXiv preprint arXiv:1909.03546*, 2019.
- [44] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North {A}merican Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Minneapolis, Minnesota: Association for

- Computational Linguistics, jun 2019, pp. 4171–4186. [Online]. Available: <https://www.aclweb.org/anthology/N19-1423>
- [45] A. Esmailzadeh and K. Taghva, "Text Classification Using Neural Network Language Model (NNLM) and BERT: An Empirical Comparison," in *Proceedings of SAI Intelligent Systems Conference*. Springer, 2021, pp. 175–189.
 - [46] S. M. Saravani, R. Banerjee, and I. Ray, "An investigation into the contribution of locally aggregated descriptors to figurative language identification," in *Proceedings of the Second Workshop on Insights from Negative Results in NLP*, 2021, pp. 103–109.
 - [47] M. Heidari and J. H. Jones, "Using bert to extract topic-independent sentiment features for social media bot detection," in *2020 11th IEEE Annual Ubiquitous Computing, Electronics Mobile Communication Conference (UEMCON)*, 2020, pp. 0542–0547.
 - [48] M. Heidari, J. H. Jones, and O. Uzuner, "Deep contextualized word embedding for text-based online user profiling to detect social bots on twitter," in *2020 International Conference on Data Mining Workshops (ICDMW)*, 2020, pp. 480–487.
 - [49] M. Heidari, J. H. J. Jones, and O. Uzuner, "An empirical study of machine learning algorithms for social media bot detection," in *2021 IEEE International IoT, Electronics and Mechatronics Conference (IEMTRONICS)*, 2021, pp. 1–5.
 - [50] M. Heidari and S. Rafatirad, "Bidirectional transformer based on online text-based information to implement convolutional neural network model for secure business investment," in *2020 IEEE International Symposium on Technology and Society (ISTAS)*, 2020, pp. 322–329.
 - [51] N. Kalantari, D. Liao, and V. G. Motti, "Characterizing the online discourse in twitter: Users' reaction to misinformation around covid-19 in twitter," in *2021 IEEE International Conference on Big Data (Big Data)*, 2021, pp. 4371–4380.
 - [52] M. Saadati, J. Nelson, A. Curtin, L. Wang, and H. Ayaz, "Application of recurrent convolutional neural networks for mental workload assessment using functional near-infrared spectroscopy," in *Advances in Neuroergonomics and Cognitive Engineering*, H. Ayaz, U. Asgher, and L. Paletta, Eds. Cham: Springer International Publishing, 2021, pp. 106–113.
 - [53] M. Heidari, S. Zad, P. Hajibabae, M. Malekzadeh, S. HekmatiAthar, O. Uzuner, and J. H. Jones, "Bert model for fake news detection based on social bot activities in the covid-19 pandemic," in *2021 IEEE 12th Annual Ubiquitous Computing, Electronics Mobile Communication Conference (UEMCON)*, 2021, pp. 0103–0109.
 - [54] M. E. Peters, M. Neumann, M. Iyyer, M. Gardner, C. Clark, K. Lee, and L. Zettlemoyer, "Deep contextualized word representations," *NAACL HLT 2018 - 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies - Proceedings of the Conference*, vol. 1, pp. 2227–2237, 2018.
 - [55] H. Zhu, I. C. Paschalidis, and A. Tahmasebi, "Clinical concept extraction with contextual word embedding," *arXiv preprint arXiv:1810.10566*, 2018.
 - [56] J. Rogers and O. Bodenreider, "SNOMED CT: Browsing the Browsers," in *Proceedings of the Third International Conference on Knowledge Representation in Medicine, Phoenix, Arizona, USA, May 31st - June 2nd, 2008*, ser. {CEUR} Workshop Proceedings, R. Cornet and K. A. Spackman, Eds., vol. 410. CEUR-WS.org, 2008. [Online]. Available: <http://ceur-ws.org/Vol-410/Paper06.pdf>
 - [57] A. E. W. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. Anthony Celi, and R. G. Mark, "MIMIC-III, a freely accessible critical care database," *Scientific Data*, vol. 3, no. 1, p. 160035, 2016. [Online]. Available: <https://doi.org/10.1038/sdata.2016.35>
 - [58] B. Yu, Z. Zhang, X. Shu, Y. Wang, T. Liu, B. Wang, and S. Li, "Joint extraction of entities and relations based on a novel decomposition strategy," *arXiv preprint arXiv:1909.04273*, 2019.
 - [59] Y. Yuan, X. Zhou, S. Pan, Q. Zhu, Z. Song, and L. Guo, "A Relation-Specific Attention Network for Joint Entity and Relation Extraction," in *IJCAI*, 2020, pp. 4054–4060.
 - [60] T. Nayak and H. T. Ng, "Effective modeling of encoder-decoder architecture for joint entity and relation extraction," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 05, 2020, pp. 8528–8535.
 - [61] L. Luo, Z. Yang, M. Cao, L. Wang, Y. Zhang, and H. Lin, "A neural network-based joint learning approach for biomedical entity and relation extraction from biomedical literature," *Journal of biomedical informatics*, vol. 103, p. 103384, 2020.
 - [62] R. Li, D. Li, J. Yang, F. Xiang, H. Ren, S. Jiang, and L. Zhang, "Joint extraction of entities and relations via an entity correlated attention neural model," *Information Sciences*, vol. 581, pp. 179–193, 2021.
 - [63] G. Bekoulis, J. Deleu, T. Demeester, and C. Develder, "Adversarial training for multi-context joint entity and relation extraction," *arXiv preprint arXiv:1808.06876*, 2018.
 - [64] J. Sun, Y. Li, Y. Shen, W. Ding, X. Shi, L. Zhang, X. Shen, and J. He, "Joint Self-Attention Based Neural Networks for Semantic Relation Extraction," *Journal of Information Hiding and Privacy Protection*, vol. 1, no. 2, p. 69, 2019.
 - [65] T.-J. Fu, P.-H. Li, and W.-Y. Ma, "GraphRel: Modeling text as relational graphs for joint entity and relation extraction," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019, pp. 1409–1418.
 - [66] K. Zhao, H. Xu, Y. Cheng, X. Li, and K. Gao, "Representation iterative fusion based on heterogeneous graph neural network for joint entity and relation extraction," *Knowledge-Based Systems*, vol. 219, p. 106888, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0950705121001519>
 - [67] S. Zheng, Y. Hao, D. Lu, H. Bao, J. Xu, H. Hao, and B. Xu, "Joint entity and relation extraction based on a hybrid neural network," *Neurocomputing*, vol. 257, pp. 59–66, 2017.
 - [68] Z. Meng, S. Tian, L. Yu, and Y. Lv, "Joint extraction of entities and relations based on character graph convolutional network and Multi-Head Self-Attention Mechanism," *Journal of Experimental & Theoretical Artificial Intelligence*, vol. 33, no. 2, pp. 349–362, 2021.
 - [69] X. Zeng, D. Zeng, S. He, K. Liu, and J. Zhao, "Extracting relational facts by an end-to-end neural model with copy mechanism," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2018, pp. 506–514.
 - [70] C. Sun, Y. Gong, Y. Wu, M. Gong, D. Jiang, M. Lan, S. Sun, and N. Duan, "Joint type inference on entities and relations via graph convolutional networks," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019, pp. 1361–1370.
 - [71] S. M. Saravani, I. Ray, and I. Ray, "Automated identification of social media bots using deep fake text detection," in *International Conference on Information Systems Security*. Springer, 2021, pp. 111–123.
 - [72] D. Dai, X. Xiao, Y. Lyu, S. Dou, Q. She, and H. Wang, "Joint extraction of entities and overlapping relations using position-attentive sequence labeling," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 33, no. 01, 2019, pp. 6300–6308.
 - [73] Y. Hong, Y. Liu, S. Yang, K. Zhang, and J. Hu, "Joint extraction of entities and relations using graph convolution over pruned dependency trees," *Neurocomputing*, vol. 411, pp. 302–312, 2020.
 - [74] A. Daigavane, B. Ravindran, and G. Aggarwal, "Understanding Convolutions on Graphs," *Distill*, vol. 6, no. 9, p. e32, 2021.
 - [75] Y. Tang, J. Yu, S. Li, Y. Tan, Q. Wu, and Others, "Span Representation Generation Method in Entity-Relation Joint Extraction," in *International Conference on Intelligent Computing*. Springer, 2021, pp. 465–476.